CHAPTER 3

Groups and Construction of Integers

3.1 Groups

We now define the concept of a group. A group is an algebraic structure, which helps us to better understand the abstract properties of the different sets of numbers defined above.

We start with what is called a binary operation, i.e. an operation such as $+, \cdot, /, -$

Definition 3.1 (Binary Operation) *Let $M \neq 0$ be a set. We call a mapping*

$$\circ\colon M \times M \longrightarrow M$$
$$(a, b) \ni M \times M \mapsto \circ(a, b) \in M$$

a binary operation on M. We usually write $a \circ b$ instead of $\circ(a, b)$.
If for the binary relation $\circ$ it holds

$$\circ(a, b) = \circ(b, a)\,, \tag{3.1}$$

we call $\circ$ commutative.

Examples of binary operations are the usual addition "$+$" or the multiplication "$\cdot$". On $M = \mathbb{N}$, $\circ = +$ is a binary operation and we write, e.g. $\circ(2, 3) = +(2, 3) = 2 + 3$.

Since the co-domain of $\circ$ is M, we call the binary operation closed.

Definition 3.2 (Group) *A group is a set $G \neq \emptyset$ equipped with a binary operation $\circ\colon G \times G \longrightarrow G$ such that we have*

Here you can find the video on construction of $\mathbb{Z}$.

For completeness, we also include the definition of a field here.

(G1) *For all $a, b, c \in G$ it holds* — Associativity

$$a \circ (b \circ c) = (a \circ b) \circ c\,. \tag{3.2}$$

(G2) *There exists $e \in G$ such that* — Neutral Element

$$e \circ a = a \circ e = a \quad \forall a \in G\,. \tag{3.3}$$

(G3) *For every $a \in G$ there exists $b \in G$ such that* — Inverse Element

$$a \circ b = b \circ a = e\,. \tag{3.4}$$

We call b the inverse element *to a and write $b = a^{-1}$.*

Definition 3.3 (Field) *Let $\mathcal{F} \neq \emptyset$ be a set equipped with the binary operations $+ \colon \mathcal{F} \times \mathcal{F} \longrightarrow \mathcal{F}$ and $\cdot \colon \mathcal{F} \times \mathcal{F} \longrightarrow \mathcal{F}$ with respective neutral elements 0 and 1. We call $\mathcal{F}$ a field if*

(F1) *$(\mathcal{F}, +)$ is a commutative (or Abelian) group.*
(F2) *$(F \setminus \{0\}, \cdot)$ is a commutative (or Abelian) group.*
(F3) *The following distributive law holds:*

$$a \cdot (b + c) = a \cdot b + a \cdot c \qquad \forall a, b, c \in \mathcal{F}. \tag{3.5}$$